

DFT Calculations Report: SCF Convergence and Pseudopotential Analysis

Fatemeh Rezaei

April 7, 2025

1 A Reminder of Definitions

1.1 What is an SCF calculation?

In my work, an SCF (Self-Consistent Field) calculation is the iterative process of solving the Kohn-Sham equations in Density Functional Theory (DFT). I update the electron density and corresponding potential until the total energy (or charge density) changes negligibly between iterations, signifying that self-consistency has been reached.

1.2 How do K-points affect speed and accuracy?

- **Accuracy:** Using a denser k-point grid (e.g., $10 \times 10 \times 10$) results in a finer sampling of the Brillouin zone and thus improves energy accuracy. In my tests with both pseudopotentials, a $6 \times 6 \times 6$ grid yielded energy differences of less than 0.001 Ry.
- **Speed:** However, increasing the k-point grid density also increases computational cost. For Aluminum, a $6 \times 6 \times 6$ grid offers an optimal balance between speed and accuracy.

1.3 What is a pseudopotential, and what role does it play?

- **Pseudopotentials** replace the core electrons with an effective potential so that the calculations can focus on the valence electrons.
- **Al-pbesol:** Uses the PBEsol functional, optimized for solid-state systems.
- **Al-rel-pbe:** A fully relativistic pseudopotential that incorporates effects such as spin-orbit coupling.
- **Energy Reference:** Each pseudopotential has its own reference energy, which is why **Al-pbesol** yields absolute total energies around -38.405 Ry while **Al-rel-pbe** gives values around -4.938 Ry. These absolute values are not directly comparable, but each set is internally consistent.

2 SCF Calculation in Quantum Espresso

2.1 Compare total energies between pseudopotentials. Are the differences significant?

- **Absolute Energies** (averages):
 - **Al-pbesol:** ~ -38.40445 Ry
 - **Al-rel-pbe:** ~ -4.93765 Ry
- **Interpretation:** The difference in absolute energies stems from the distinct energy references in the pseudopotentials. What is important is that both exhibit consistent convergence behavior across different `ecutwfc` values.

2.2 How does `ecutwfc` effect energy?

The parameter `ecutwfc` sets the kinetic energy cutoff for the plane-wave basis set. Increasing `ecutwfc` expands the basis, improving the accuracy of the wavefunction representation. I tested with `ecutwfc` = 40, 60, and 100 Ry. For both pseudopotentials:

- At `ecutwfc` = 40 Ry, the total energies were:
 - `Al-pbesol`: -38.40549, -38.40186, -38.40554, and -38.40487 Ry.
 - `Al-rel-pbe`: -4.93865, -4.93517, -4.93867, and -4.93807 Ry.
- At `ecutwfc` = 60 Ry, the energies were nearly identical:
 - `Al-pbesol`: -38.40550, -38.40187, -38.40555, and -38.40488 Ry.
 - `Al-rel-pbe`: -4.93866, -4.93518, -4.93868, and -4.93808 Ry.
- At `ecutwfc` = 100 Ry, the results were again nearly the same:
 - `Al-pbesol`: -38.40550, -38.40187, -38.40556, and -38.40488 Ry.
 - `Al-rel-pbe`: -4.93867, -4.93518, -4.93869, and -4.93809 Ry.

The variations in total energy across the different `ecutwfc` values are negligible (less than 0.005 Ry), confirming that my basis set is sufficiently complete.

Table 1: Convergence Analysis for Different `ecutwfc` Values (Averages)

Pseudopotential	k-grid	Total Energy (Ry)		
		<code>ecutwfc</code> = 40	<code>ecutwfc</code> = 60	<code>ecutwfc</code> = 100
<code>Al-pbesol</code>	$6 \times 6 \times 6$	-38.40445	-38.40445	-38.40445
<code>Al-rel-pbe</code>	$6 \times 6 \times 6$	-4.93765	-4.93765	-4.93765

3 Key Conclusions

- A $6 \times 6 \times 6$ k-point grid provides a good compromise between accuracy and computational cost for Aluminum.
- Although the absolute energies differ between the pseudopotentials due to their distinct energy references, both show excellent convergence.
- The tests with `ecutwfc` = 40, 60, and 100 Ry reveal that total energy variations are negligible (less than 0.005 Ry), confirming that the chosen cutoff is sufficient.
- These convergence tests underscore the importance of systematically checking both k-points and plane-wave cutoffs to ensure reliable DFT results.